

Ming Li

PhD Candidate, Computer Science, University of Maryland

Expected Spring 2027

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2026 Apple Scholar in AI/ML | 2× ACL Oral (2025, 2026) | ACL'26 SAC Highlight

I study how to train, understand, and rigorously evaluate LLMs and agentic systems, with a focus on post-training, reasoning behavior, and reliable AI evaluation.

Selected Research

Data-Centric LLM Post-Training

Data selection and curation for efficient LLM post-training, with gradient-based analysis of data quality.

- **IFD / Cherry LLM** — [NAACL'24](#). Self-guided data selection achieved competitive instruction tuning with 10% of the data in Alpaca/WizardLM experiments; **470+ citations**.
- **Data Quality through Layer-wise Gradients** — [ACL'26](#). Used layer-wise gradient spectra to explain data-quality metrics across instruction and reasoning post-training. **AC-recommended Oral • Top-15% meta-review**.

Also: Superfiltering ([ACL'24](#)), ATLaS ([ACL'25 Findings](#); tuning on 30% of trajectory steps), Reflection-Tuning ([ACL'24 Findings](#)). **1K+ GitHub stars** across these post-training codebases.

Reasoning & Model Behavior

LLM interpretability and model-behavior analysis through gradients, token dynamics, and reasoning traces.

- **Schoenfeld's Anatomy / ThinkARM** — **ACL'26 Oral & SAC Highlight**. Functional decomposition of reasoning traces reveals how efficiency interventions alter reasoning structure.
- **Fast vs. Slow Thinking** — **ACL'25 Oral**. Layer-wise gradient analysis characterizes how direct-answer and explicit-reasoning training differ.

Also: Missing Premise / Overthinking ([COLM'25](#); **16K+ alphaXiv views**); visual reasoning RL analysis ([COLM'26](#)).

AI Evaluation & Reliability

Stress-testing models and agents to uncover capability limits, behavioral failures, and unreliable judgments.

- **Moltbook / AI Agent Society** — [ACM CAIS'26](#). Diagnosed multi-agent interaction dynamics, finding persistent individual inertia without stable collective consensus; **Science News interview**.
- **ColorBench** — [NeurIPS'25 Datasets & Benchmarks](#). First comprehensive VLM color benchmark: 5.8K+ questions across 11 perception, reasoning, and robustness tasks; testing 32 VLMs reveals weak scaling and misleading color cues.

Also: RobustReview / SciCore (2026 manuscript; research lead); ClawEnvKit (co-lead; 1,040 agent environments).

Industry & Visiting Research Experience

MBZUAI — Visiting Researcher

Feb–May 2026

Rhetoric Reward-Hacking (2026 preprint): Built and ran a **large-scale LLM rewriting-and-evaluation pipeline over 4,200 full manuscripts and 42K+ AI reviews**, orchestrating multiple LLM APIs for controlled, recursive, and reviewer-guided interventions while enforcing content-preservation constraints; exposed LLM-as-a-judge sensitivity to evidence framing and novelty claims.

Amazon — Applied Scientist Intern

May–Aug 2025

RLAAR ([ACL'26](#)): Developed **curriculum reinforcement learning with verifiable rewards (RLVR)** for multi-turn reliability by **extending verl to support fully dynamic multi-turn on-policy rollouts under GRPO** and implementing mixed verifiable accuracy/abstention rewards; improved Qwen3-8B Lost-in-Conversation retention from 62.6% to 75.1%.

Microsoft Research — Research Intern

Feb–May 2025

FoReal-Decoding ([NeurIPS'25 Efficient Reasoning Workshop](#)): Token-level analysis revealed reasoning/non-reasoning LLM divergence at sentence-opening “thinking cues.” Implemented a **custom collaborative decoder over Hugging Face Transformers** for efficient inference, reducing theoretical FLOPs by 30–50% and chain-of-thought (CoT) length by up to 40%.

Adobe Research — Research Scientist/Engineer Intern

May–Nov 2024

TRIG ([ICLR'25 Reasoning and Planning for LLMs Workshop](#)): Introduced the first benchmark and instruction-tuning dataset for text-rich document QA grounding. **Built a synthetic data pipeline from raw documents to ~96K training examples via multimodal alignment, iterative refinement, and filtering**; implemented distributed SFT of LLaVA-13B and MLLM evaluation.

Selected Publications

Selected publications and manuscripts · 20+ first/co-first-author papers overall

* Equal contribution

Full list: [Homepage](#) | [Google Scholar](#)

Data-Centric LLM Post-Training

From Quantity to Quality: Boosting LLM Performance with Self-Guided Data Selection for Instruction Tuning.

M. Li, Y. Zhang, Z. Li, et al. — [NAACL 2024](#)

Superfiltering: Weak-to-Strong Data Filtering for Fast Instruction-Tuning.

M. Li, Y. Zhang, S. He, et al. — [ACL 2024](#)

How Instruction and Reasoning Data Shape Post-Training: Data Quality through the Lens of Layer-wise Gradients.

M. Li, Y. Li, Z. Li, T. Zhou — [ACL 2026](#); **AC-recommended Oral · Top-15% meta-review**

Mitigating Lost in Multi-turn Conversation via Curriculum RL with Verifiable Accuracy and Abstention Rewards.

M. Li, P. Chen, Z. Zhang, et al. — [ACL 2026](#)

ATLaS: Agent Tuning via Learning Critical Steps.

Z. Chen*, M. Li*, Y. Huang, et al. — [Findings of ACL 2025](#)

Reasoning & Model Behavior

Schoenfeld’s Anatomy of Mathematical Reasoning by Language Models.

M. Li*, C. Fan*, Y. Cheng*, S. Feizi, T. Zhou — **ACL 2026 Oral & SAC Highlight**

What Happened in LLMs Layers when Trained for Fast vs. Slow Thinking: A Gradient Perspective.

M. Li, Y. Li, T. Zhou — **ACL 2025 Oral**

Missing Premise Exacerbates Overthinking: Are Reasoning Models Losing Critical Thinking Skill?

C. Fan*, M. Li*, L. Sun, T. Zhou — [COLM 2025](#)

What Does RL Improve for Visual Reasoning? A Frankenstein-Style Analysis.

X. Li*, M. Li*, T. Zhou — [COLM 2026](#)

AI Evaluation & Reliability

Does Socialization Emerge in AI Agent Society? A Case Study of Moltbook.

M. Li*, X. Li*, T. Zhou — [ACM CAIS 2026](#)

How Can Rhetoric Reward-Hack AI Reviewers? Dissecting Rhetorical Sensitivity in AI-Based Peer Review.

M. Li*, C. Wang*, X. Li, et al. — 2026 preprint

ColorBench: Can VLMs See and Understand the Colorful World? A Comprehensive Benchmark for Color Perception, Reasoning, and Robustness.

Y. Liang*, M. Li*, C. Fan, et al. — [NeurIPS 2025 Datasets & Benchmarks](#)

BenTo: Benchmark Reduction with In-Context Transferability.

H. Zhao, M. Li, L. Sun, T. Zhou — [ICLR 2025](#)

Education

University of Maryland — PhD Candidate, Computer Science

2023–Present

Advisor: Tianyi Zhou | Expected Spring 2027

Texas A&M University — MS, Computer Science

2021–2023

Advisor: Ruihong Huang

Xi’an Jiaotong University — BS, Computer Science

2016–2020

Research Leadership & Recognition

Recognition. **Best Research Award, e-Assessment Awards 2026** — project lead & first author ([EPM’25](#)) ·

UMD Graduate School **Dean’s Fellowship** (2023).

Community & Outreach. ACL ARR Area Chair · AI/ML technical content creator, **2.8K followers** on RedNote (Xiaohongshu) · [CIKM’25](#) tutorial contributor.

Mentorship. Mentored junior researchers across multiple first-author projects, from problem formulation through publication and independent ownership.

Technical Skills. PyTorch · Hugging Face Transformers · verl · vLLM · distributed training (SFT) · RLHF/RLVR · PPO · DPO · GRPO · on-policy distillation (OPD) · reward design · synthetic data · multimodal learning · LLM/VLM evaluation.